SECTION 2: MATHEMATICS PROFICIENCY
Time -- 45 minutes
50 Questions (61 – 110)

NAME: _____
SCHOOL: _____
SIMULATED EXAM – SET B

For each question, decide which the best answer choice is, and darken the corresponding oval on the answer sheet.

61. If θ is an angle in quadrant II, which of the following is negative?
A. $cos\theta + 1$
B. $sec\theta csc\theta$
C. $sin^3\theta$
D. $cot\theta tan\theta$
E. $cos^2\theta$

62. Simplify: $\frac{1}{cot\theta + tan\theta}$
A. $sec\theta csc\theta$
B. $sin\theta cos\theta$
C. $\frac{1}{sec\theta csc\theta}$
D. 1
E. -1

63. Find the exact value of $(\sec 60° + \tan 45°)$.
A. $2 + \sqrt{2}$
B. $\sqrt{2} + 0.5$
C. $\sqrt{3} + 2$
D. 3
E. 2

64. If θ is an acute angle, which of the following is equal to $\tan(180 - \theta)$?
A. $\tan\theta$
B. $1 + \tan\theta$
C. $1 - \tan\theta$
D. $-\tan\theta$
E. undefined

65. The domain of the trigonometric function $y = \tan 3x$ is the set of all real numbers except
A. $\ldots, -\pi, -\frac{2\pi}{3}, -\frac{\pi}{3}, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \ldots$
B. $\ldots, -\frac{\pi}{2}, -\frac{\pi}{3}, -\frac{\pi}{6}, \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}, \ldots$
C. $\ldots, -\frac{5\pi}{6}, -\frac{\pi}{2}, -\frac{\pi}{6}, \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \ldots$
D. $\ldots, -\frac{5\pi}{2}, -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$
E. $\ldots, -\frac{3\pi}{2}, -\pi, -\frac{\pi}{2}, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \ldots$

66. Find the period of the trigonometric function $y = 2\cos\left(3x - \frac{\pi}{2}\right)$.
A. $\frac{2\pi}{3}$
B. $\frac{\pi}{3}$
C. 2π
D. 4π
E. $\frac{\pi}{4}$

67. If x is an angle in standard position and $\cos x = \frac{3}{4}$ and $\sin x < 0$, find the exact value of $\tan x$.
A. $-\frac{\sqrt{7}}{3}$
B. $-\frac{5}{3}$
C. $-\frac{4}{3}$
D. $\frac{5}{3}$
E. $\frac{\sqrt{7}}{3}$

68. A pair of dice is rolled. What is the probability of getting a sum of at least five?
A. $\frac{5}{6}$
B. $\frac{1}{6}$
C. $\frac{13}{18}$
D. $\frac{5}{18}$
E. $\frac{8}{9}$

69. A pair of dice is rolled. What is the probability of getting a sum of at most 10?
A. $\frac{5}{6}$
B. $\frac{1}{12}$
C. $\frac{11}{12}$
D. $\frac{1}{18}$
E. $\frac{8}{9}$

70. Three coins are tossed. What is the probability of getting at most two heads?
A. $\frac{1}{8}$
B. $\frac{3}{8}$
C. $\frac{5}{8}$
D. $\frac{7}{8}$
E. $\frac{1}{2}$

71. Two balls are drawn in succession without replacement from a bag containing 8 red, 7 blue, and 10 yellow marbles. If the first ball drawn is blue, then what is the probability that the second marble drawn is red?
A. $\frac{1}{3}$
B. $\frac{1}{2}$
C. $\frac{8}{25}$
D. $\frac{1}{4}$
E. $\frac{13}{24}$

72. The table below shows the scores of 5 students.

Student	Scores
A	85
B	89
C	78
D	85
E	88

Which of the following is true?
A. mean = median = mode
B. mean > median > mode
C. median > mode > mean
D. median > mean > mode
E. mode > median > mean

73. Mr. James, a History teacher, computed the mean grade of his 10 students in a certain class. If he would give each student a bonus of 2 points, what would happen to the mean grade of the 10 students?
A. The mean grade would increase by 20 points.
B. The mean grade would increase by 2 points.
C. The mean grade would stay the same.
D. The mean grade would increase by 10 points.
E. There was no way of knowing exactly how many points the mean increased.

74. Andrea computed the mean of her 8 scores in Math. After she had computed the mean, her teacher gave a bonus of 24 points to be added on her total score. What was the effect of these additional points on Andrea's mean score?
    A. The mean increased by 12 points.
    B. The mean remained the same.
    C. The mean increased by 3 points.
    D. The mean increased thrice,
    E. There was no way of knowing exactly how many points the mean increased.

75. A statistics student computed the standard deviation of 25 scores and he found it to be equal to 4. If 10 were subtracted from each score, what would be the new standard deviation?
    A. 2.5
    B. 0
    C. 3.25
    D. 6.5
    E. None of these

76. Eight different Chemistry books and ten different Physics books are to be arranged in a shelf. In how many ways can such books be arranged if books of the same subject shall be placed side by side?
    A. $18!$
    B. $8!\,10!$
    C. $8! + 10!$
    D. $2!\,8!\,10!$
    E. $2!\,18!$

77. The sum of the present ages of Alex and Ben is 39 years. In 5 years, Alex will be thrice as old as Ben was 4 years ago. How old will Ben be ten years from now?
    A. 14 years old
    B. 24 years old
    C. 25 years old
    D. 32 years old
    E. 35 years old

78. The present age of a mother is equal to the sum of the ages of her triplets. Thirty nine years hence, the sum of the ages of the triplets will be twice the age of their mother. Find the present age of the mother.
    A. 32 years old
    B. 39 years old
    C. 41 years old
    D. 43 years old
    E. years old

79. A mother was 27 years old when her eldest child was born. Ten years from now, the sum of the ages of the mother and her eldest child will be equal to 57 years. How old is the mother now?
    A. 27 years old
    B. 32 years old
    C. 42 years old
    D. 45 years old
    E. 52 years old

80. Two cargo vessels left the same port at the same time and traveled in opposite directions. If the speed of the two vessels were 52 kph and 43 kph, how long would it take them to be 190 km apart?
    A. 1 hour
    B. 1.5 hours
    C. 2 hours
    D. 2.25 hours
    E. 2.5 hours

81. Maria, a laboratory assistant, receives a request of 400 milliliters of a 50% denatured alcohol solution, but she only has 65% and 35% denatured alcohol solutions on hand. In order to produce the request, how many milliliters of 65% and 35% alcohol solutions should she mix?
   A. 100 mL and 300 mL, respectively
   B. 150 mL and 250 mL, respectively
   C. 200 mL of each
   D. 250 mL and 150 mL, respectively
   E. 300 mL and 100 mL, respectively

82. Mon and Jacky together can do a job in 4 hours. If Mon can finish job in 9 hours, how long would it take Jacky to finish the job by herself?
   A. 5 hours
   B. 7 hours
   C. 7.1 hours
   D. 7.2 hours
   E. 8 hours

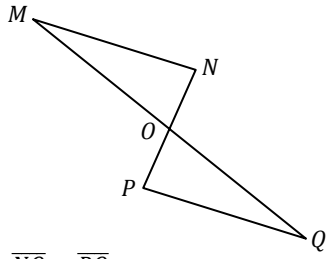
83. A rectangular lot has an area of 32 square meters. Its perimeter is 24 meters. Find the dimensions of the rectangular lot.
   A. 4 meters by 8 meters
   B. 3 meters by 9 meters
   C. 2 meters by 16 meters
   D. 1 meter by 32 meters
   E. 3 meters by 7 meters

84. Two numbers differ by 13. When the smaller number is doubled and the larger is increased by 7, their sum would be 59. What is the larger number?
   A. 12
   B. 13
   C. 18
   D. 21
   E. 26

85. A man invested certain amounts in two different accounts. The first account earned 6% simple annual interest and the second account earned 4.5% simple annual interest. The amount the man invested in the first account was $6,000 more than the second account. After a year, the first account earned twice as much as the second account. How much did the man invest in each account?
   A. $10,000 at 6% and $4,000 at 4.5%
   B. $12,000 at 6% and $18,000 at 4.5%
   C. $15,000 at 6% and $9,000 at 4.5%
   D. $9,000 at 6% and $15,000 at 4.5%
   E. $18,000 at 6% and $12,000 at 4.5%

86. Which of the following is not true?
   A. If two angles are congruent and complementary, then the measure of each angle is $45°$.
   B. If two angles are congruent and supplementary, then each angle is a right angle.
   C. Vertical angles are congruent.
   D. If two angles are supplementary, then the angles form a linear pair.
   E. All right angles are congruent.

87. Refer to the figure below. Which of the following will make $\Delta MNO \cong \Delta QPO$ by $SAS\ Postulate$?

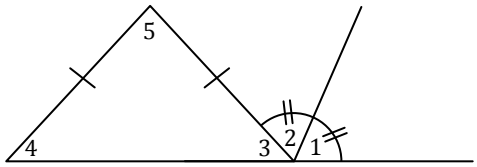


   A. $\angle N \cong \angle P$; $\overline{NO} \cong \overline{PO}$
   B. $\overline{MN} \cong \overline{QP}$; $\overline{MN} // \overline{QP}$
   C. $\overline{MQ}$ and $\overline{NP}$ bisects each other.
   D. $\overline{MN} // \overline{QP}$
   E. O is the midpoint of $\overline{MQ}$.

88. The length of the side of an equilateral triangle is 6 cm. Find the altitude of the triangle.
A. 3
B. $6\sqrt{3}$
C. $3\sqrt{2}$
D. $3\sqrt{3}$
E. 12

89. What is the area of the parallelogram if the base and the altitude were both equal to $2x+3$?
A. $4x^2+12x+9$
B. $\frac{1}{2}(4x^2+12x+9)$
C. $4x^2-9$
D. $\frac{1}{2}(4x^2-9)$
E. $4x^2+9$

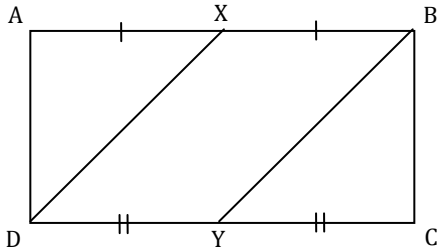
90. Refer to the figure below.



Which of the following is false?
A. $m\angle 3 = m\angle 4$
B. $m\angle 4 + m\angle 5 = m\angle 1 + m\angle 2$
C. $m\angle 1 = m\angle 4$
D. $2m\angle 1 + m\angle 3 = 180°$
E. $2m\angle 1 = m\angle 4 + m\angle 5$

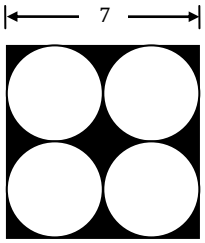
91. The length of the longest side of a right triangle is 50. If one of the acute angles of the right triangle is $30°$, find the length of the other two sides.
A. 30 and 40
B. $25\sqrt{3}$ and 30
C. 30 and 60
D. 25 and $25\sqrt{3}$
E. 25 and $25\sqrt{2}$

92. Refer to the figure below.



Which of the following is true about the area of rectangle $ABCD$ and parallelogram $DXBY$?
A. $Area_{ABCD} = \frac{1}{2}Area_{DXBY}$
B. $Area_{ABCD} = \frac{1}{4}Area_{DXBY}$
C. $Area_{ABCD} = 2Area_{DXBY}$
D. $Area_{ABCD} = 3Area_{DXBY}$
E. $Area_{ABCD} = 4Area_{DXBY}$

93. Find the area of the shaded region.



A. $49(1-\pi)$
B. $14(4-\pi)$
C. $49\left(1-\frac{\pi}{4}\right)$
D. $49-7\pi$
E. $49-\frac{\pi}{4}$

94. In a parallelogram, which of the following is always true?
A. Adjacent sides are congruent.
B. Opposite angles are supplementary.
C. Diagonals are congruent.
D. Opposite sides are perpendicular.
E. Adjacent angles are supplementary.

95. Find the area of a sector of a circle whose radius is 5 units and whose central angle is 54°.

A. $\frac{5\pi}{2}$

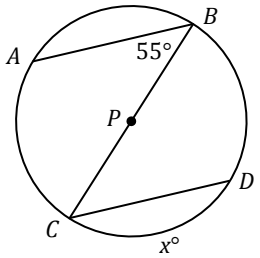
B. $\frac{15\pi}{4}$

C. $\frac{20\pi}{3}$

D. $\frac{18\pi}{5}$

E. $\frac{25\pi}{4}$

96. In circle P, BC is a diameter and AB and CD are parallel chords. Find the value of x.



A. 125°
B. 70°
C. 110°
D. 75°
E. 95°

97. $7\sqrt[3]{56} - 7\sqrt{56} - \sqrt[3]{189} + \sqrt{126} =$

A. 0

B. $11\sqrt{14} - 11\sqrt[3]{7}$

C. $11\left(\sqrt[3]{7} - \sqrt{14}\right)$

D. $29\sqrt[3]{7} - 24\sqrt{14}$

E. $-\sqrt{7}$

98. Which of the following shows additive identity property?

A. $-ab + ab = 0$
B. $0 + (-ab) = -ab$
C. $0 + (-ab) = (-ab) + 0$
D. $(1)(-ab) = -ab$
E. $(c)(-ab) = (-ab)(c)$

99. What value(s) of k could make the expression below a perfect square trinomial?

$$x^2 - \frac{3}{2}x + k$$

A. 9

B. $\frac{9}{4}$

C. $\frac{9}{16}$

D. $\frac{1}{9}$

E. No values of k could make the expression a perfect square trinomial.

100. Solve for x:

$$49^{2x-1} = \left(\frac{1}{7}\right)^{-2}$$

A. $x = 1$
B. $x = 0$
C. $x = -3$
D. $x = -1$
E. $x = 4$

101. Which of the following is a factor of $8x^3 + 27y^3$?

A. $2x - 3y$

B. $4x^2 - 6xy + 9y^2$

C. $4x^2 - 12xy + 9y^2$

D. $4x^2 + 6xy + 9y^2$

E. $8x + 27y$

102. What is the square of $4x - 2y - z + 4$?

A. $16x^2 + 4y^2 + z^2 + 16$
B. $16x^2 + 4y^2 + z^2 + 16 - 4xy - 4xz - 16x + 2yz - 8y - 4z$
C. $16x^2 - 4y^2 - z^2 + 16 - 8xy - 8xz - 32x + 4yz - 16y - 8z$
D. $16x^2 + 4y^2 + z^2 + 16 - 16xy - 8xz + 32x + 4yz - 16y - 8z$
E. $16x^2 - 4y^2 - z^2 + 16$

103. The volume of a 10-cm high cylinder is 640π cm³. What is the radius of the base of the cylinder?
A. 4 cm
B. 6 cm
C. 8 cm
D. 12 cm
E. 16 cm

104. The sum and product of the roots of a quadratic equation are -5 and $\frac{3}{4}$, respectively. Find the quadratic equation.
A. $4x^2 - 20x + 3 = 0$
B. $4x^2 + 20x + 3 = 0$
C. $4x^2 + 5x + 3 = 0$
D. $4x^2 - 5x + 3 = 0$
E. $4x^2 - 5x - 3 = 0$

105. Solve for x: $\log_{2x-3} 9 = \frac{1}{2}$
A. 3
B. 7
C. 24
D. 42
E. 50

106. The roots of the $(x-3)(x+1) = 2$ are
A. rational and unequal.
B. rational and equal.
C. irrational and unequal.
D. irrational and equal.
E. imaginary.

107. Determine the greatest integer in the solution set of the inequality $3(x-4) - 5(4x-1) > 10$.
A. -8
B. -1
C. 0
D. 3
E. -2

108. The domain of the function $f(x) = \frac{x}{x^2+1}$ is
A. the set of all real numbers except 1 and -1
B. the set of all real numbers
C. the set of all real numbers except -1
D. the set of all real number greater than 1
E. the set of all real number greater than 0

109. If x varies directly as the square of y and inversely as the square root of z, and $x = 60$ when $y = 6$ and $z = 81$, find x when $y = 8$ and $z = 144$.
A. 45
B. 60
C. 75
D. 80
E. 90

110. Find the equation of the line perpendicular to the line $y = 4 - 3x$ and whose $x - intercept$ is -7.
A. $x - 3y = 21$
B. $3x + y = -7$
C. $x - 3y = -7$
D. $x + 3y = 21$
E. $x - 3y = 7$